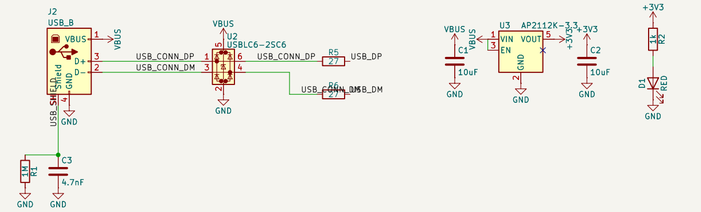
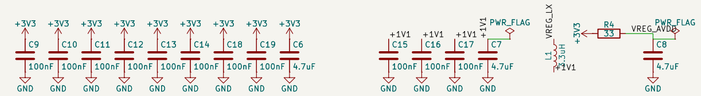
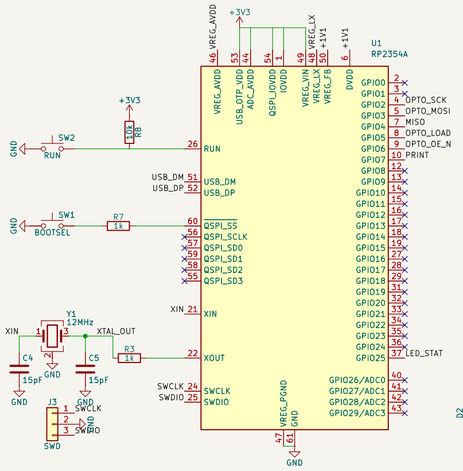
SOLARTRON 7075 DVM — ISOLATED USB INTERFACE (RP2354A)

USB SIDE (GND / +3V3 / VBUS)



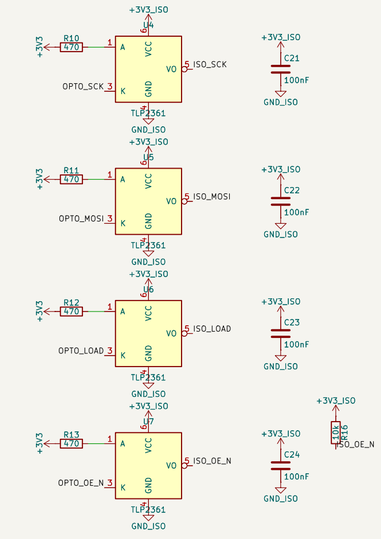
J2: TE 5787834-1 vertical USB-B on the TOP side (away from the meter).
RP2354A: internal 2 MB flash, QSPI data pins NC; BOOTSEL on QSPI_SS via R7.
L1 must be the polarity-marked AOTA-B201610S3R3 — place per RP2350 layout guide.



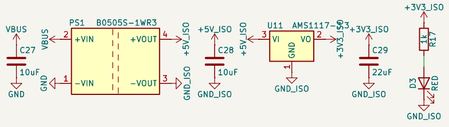


ISOLATION BARRIER

Only U4–U9 (TLP2361, 3.75 kVrms) and PS1 (B0505S, 1.5 kV) may bridge GND and GND_ISO.
PCB: keep a continuous copper-free gap (>= 5 mm) under the optos and PS1; no Y-capacitor fitted.
All opto links are net non-inverting (LED sinks from the driver; TLP2361 output inverts).
Unpowered / reset driver = LED off = receiver output HIGH.



ISOLATED SUPPLY: B0505S (unregulated, up to ~6 V at light load) -> AMS1117-3.3 -> +3V3_ISO



SERIAL LINK ACROSS THE BARRIER (SPI mode 3, <= 1 MHz; PIO or SPIO)

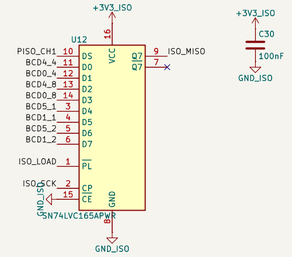
GPIO2 SCK -> U4 -> ISO_SCK GPIO3 MOSI -> U5 -> ISO_MOSI GPIO5 LOAD -> U6 -> ISO_LOAD
GPIO6 OE_N -> U7 -> ISO_OE_N GPIO4 MISO <- U8 <- '165 chain GPIO7 PRINT <- U9 <- DVM pin 34 (IRQ)

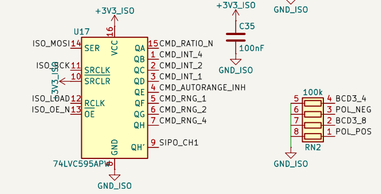
ISO_LOAD low : all five '165s capture the 36 DVM outputs (asynchronous parallel load).
ISO_LOAD rising: both '595s latch the command word shifted in during the previous transfer.
Transfer = pulse LOAD, then 56 SCK clocks:
  MISO bits 1–40 : U12 D7..D0, U13 D7..D0, U14, U15, U16 D7..D0 (U16 D3..D0 = 0,0,1,1 signature)
  MISO bits 41–56: loop-back of the previous 16-bit command word (U18 QH' -> U16 DS)
  MOSI bits 41–56: new command word (first bit ends in U18 QH, last bit in U17 QA)
  Always resend the full command word — every LOAD edge re-latches the '595s.
ISO_OE_N is held high (R16 + opto idle) until firmware drives GPIO6 low after writing a valid word,
so with USB unplugged or the MCU in reset the DVM inputs float to their own defaults (front panel usable).

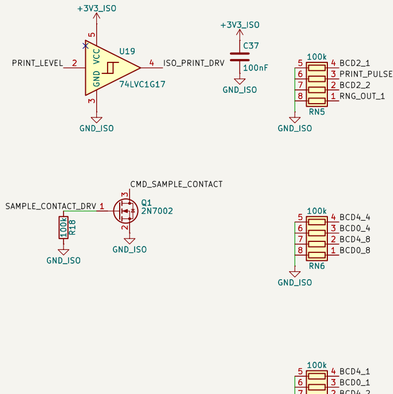
Meter-side logic is 74LVC at 3.3 V: inputs 5.5 V tolerant, V_IH 2.0 V (DVM high >= 2.4 V via 6k),
loff on all outputs — the DVM is never back-powered and pin 38 (FRONT PANEL LOCKOUT) is never pulled
low while this board is unpowered. RN1–RN9 (100k) keep the '165 inputs defined if the DVM is off.
SAMPLE: contact closure on pin 39 via Q1 (U18 QA high = closed), or >100 us pulse on pin 40 (U18 QC, 3.3 V).
NOTE: the Section 9 tables disagree on whether pin 42 or 43 is the AC-select bit — confirm on the instrument.

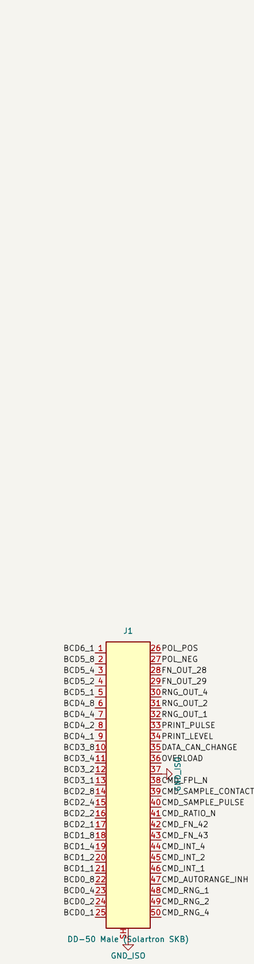
Pin-swapped bit map (MISO bit 1 first; each '165 D7..D0, DD-50 pin in brackets):
  U12: BCD1_2(20) BCD5_2(4) BCD1_1(21) BCD5_1(5) BCD0_8(22) BCD4_8(6) BCD0_4(23) BCD4_4(7)
  U13: PRINT_LEVEL(34) BCD6_1(1) BCD1_8(18) DATA_CAN_CHANGE(35) BCD5_8(2) BCD1_4(19) OVERLOAD(36) BCD5_4(3)
  U14: POL_POS(26) BCD3_8(10) POL_NEG(27) BCD3_4(11) FN_OUT_28(28) BCD3_2(12) FN_OUT_29(29) BCD3_1(13)
  U15: RNG_OUT_4(50) BCD2_8(14) RNG_OUT_2(31) BCD2_4(15) RNG_OUT_1(32) BCD2_2(16) PRINT_PULSE(33) BCD2_1(17)
  U16: BCD0_2(24) BCD4_2(8) BCD0_1(25) BCD4_1(9) then D3..D0 = 0,0,1,1 signature
  U18 QA..QE: SAMPLE_CONTACT_DRV CMD_FPL_N(38) CMD_SAMPLE_PULSE(40) CMD_FN_42(42) CMD_FN_43(43); QF–QH unused
  U17 QA..QH: CMD_RATIO_N(41) CMD_INT_4(44) CMD_INT_2(45) CMD_INT_1(46) CMD_AUTORANGE_INH(47) CMD_RNG_1(48) CMD_RNG_2(49) CMD_RNG_4(50)

METER SIDE (GND_ISO = DD-50 pin 37 EARTH / logic 0, +3V3_ISO)









J1: CAX DP-G-50P-ZC-YCLS vertical DD-50 male (LCSC C54573066), mounted on the BOTTOM side, facing the meter; mates with the 50-way systems-interface socket (Section 9 SKB pin-out). Board is held by 4-40 UNC jackscrews through the 3.3 mm holes on 61.0 mm centres. Shell (SH) tied to GND_ISO (pin 37 is EARTH in the 7075).

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